

WORKPACK 3

Basic Cleaning and Transformation

Level 3 Data Technician · GreenWheels Urban Mobility · Self-Diagnostic Pack

Get Ready to Smash This

A head-start check — this is not a test

Below are eight things this pack covers. Tick the ones you can already do.

If you can tick all eight: jump straight to the Mini-Mission on page 8 and prove it. Then move on to Pack 4.

- ☐ I know how to open a dataset in Power Query using Data → Get Data.
- ☐ I can rename a column in Power Query by double-clicking the column header.
- ☐ I can change a column's data type using the data type icon on the left of the header.
- ☐ I can use Trim and Clean to remove unwanted spaces from text values.
- ☐ I can use Replace Values to fix an incorrect text value across an entire column.
- ☐ I can use Split Column → By Delimiter to separate two pieces of information in one column.
- ☐ I can use Fill Down to copy a value down through blank cells below it.
- ☐ I know how to Remove Duplicates and filter out null rows in Power Query.

Priya at the team stand-up says: "Marcus just loaded the data straight into a pivot table. I don't think he cleaned it first." Let's find out why that matters.

What This Pack Is About

Power Query is a tool inside Excel that lets you clean, reshape, and transform data before it reaches your spreadsheet. The changes you make in Power Query do not touch the original file. Everything is recorded as a set of steps that you can run again on any updated version of the data.

This pack uses Pack3_GW_data.xlsx, a version of the GreenWheels dataset that contains several realistic data quality issues. You will fix each one using a different Power Query technique. By the end, you will have a clean, correctly typed dataset ready for analysis.

What Is in the File

The file has 42 records. When you open it in Power Query you will notice:

NOTE

Some Station Name values have extra spaces at the start or end: ' Central Square ' appears instead of 'Central Square'.

Some Zone values are in the wrong case: 'north', 'EAST', and 'west' appear instead of 'North', 'East', and 'West'.

Some Period values use a slash: 'Q1/2024' instead of 'Q1 2024'. These need splitting into a Quarter column and a Year column.

Three consecutive North Bridge records have blank Zone values. They need filling from the record above.

HC-041 is a duplicate of HC-007. HC-042 has four blank numeric fields. Both need removing.

The Applied Steps Panel

Every action you take in Power Query appears in the Applied Steps panel on the right side of the screen. This is the full cleaning journey you will build during this pack:

1	Source — connect to the workbook
2	Navigation — select the data sheet
3	Changed Type — set correct data types
4	Trimmed Text — remove leading/trailing spaces
5	Replaced Values — fix zone capitalisation
6	Split Column — separate Period into Quarter and Year
7	Fill Down — fill blank Zone cells
8	Removed Duplicates — delete HC-041
9	Filtered Rows — remove HC-042 (blank record)
10	Close & Load — send clean data to the sheet

The Skills

Work through each chunk in order. Use Pack3_GW_data.xlsx throughout. Keep the Applied Steps panel visible as you go — by the end, it should match the list above.

1

Connect to the Data

Opening Power Query for the first time

SCENARIO

Priya has sent you Pack3_GW_data.xlsx.

"Before you do anything — load it into Power Query. Do not just open it and start filtering. We want every step recorded."

Power Query sits inside Excel. You open it once, connect it to a file, and from then on all your cleaning steps are saved there. The original data file is never changed.

WORD TO KNOW

Power Query

A tool built into Excel (Data tab → Get Data) that lets you connect to a data source, clean and reshape the data, and load a clean copy into your spreadsheet. Every step is recorded and can be re-run.

WORD TO KNOW

Applied Steps

The list of steps on the right side of the Power Query Editor. Each time you make a change, a new step is added. You can click any step to see the data at that point.

To load the file: open a blank Excel workbook. Go to the Data tab → Get Data → From File → From Workbook. Browse to Pack3_GW_data.xlsx and click Import. In the Navigator window, tick the data sheet and click Transform Data. The Power Query Editor opens.

TRY IT

Open Pack3_GW_data.xlsx using Data → Get Data → From File → From Workbook. In the Navigator, select the sheet called '2024 Data (Needs Cleaning)' and click Transform Data.

Look at the Applied Steps panel on the right. How many steps are already there? What are they called?

CHECK YOURSELF

Why do we use Power Query rather than editing the Excel file directly?

Answer: Power Query keeps the original file unchanged and records every step. If the data is updated, you can refresh and all steps run again automatically. Direct edits cannot be repeated or audited.

2

Rename Columns and Set Data Types

Getting the structure right before cleaning

SCENARIO

"The column header says Total Revenue (£) — that bracket and pound sign will cause problems in formulas later.

Also, Period Start Date is loading as text. It needs to be a proper date type." — Marcus, reading from the style guide.

The first thing to fix in any dataset is the structure: make sure column names are clean and every column has the right data type. If a number column is typed as text, you cannot do maths on it.

WORD TO KNOW

Data type

The kind of data a column holds: whole number, decimal, text, date, true/false. Power Query shows a small icon on the left of each column header — a 1.2 icon means decimal, ABC means text, a calendar icon means date.

WORD TO KNOW

Rename column

Double-click any column header in Power Query to edit its name. The rename is recorded as a step called 'Renamed Columns'.

TRY IT

Double-click the 'Total Revenue (£)' column header. Rename it to 'Total Revenue'. Press Enter.

Click the data type icon (the small symbol on the left of the 'Period Start Date' column header). Change it from Text (ABC) to Date. Confirm the change when prompted.

Check the Total Hires, Avg Hire Duration, Total Revenue, Active Bikes, and Customer Satisfaction Score columns. Are they all set to a number type (Whole Number or Decimal Number)? Fix any that are still Text.

CHECK YOURSELF

What happens if you try to calculate the average of a column that is typed as Text in Power Query?

Answer: Power Query will not allow number calculations on a text column. The column would show 'ABC' in the header and any sum or average would return an error or simply not be available.

3

Trim Text and Replace Values

Fixing spaces and wrong capitalisation

SCENARIO

Brian has run a summary table by Zone. He notices "north" and "North" are showing as two separate zones.

"It looked fine in the raw data — I could not see the spaces in Station Name either until the COUNTIF returned zero." How do you find and fix invisible issues like these?

Extra spaces and inconsistent capitalisation are two of the most common data quality problems. You cannot see them by reading the data, but they break every formula that tries to match text.

WORD TO KNOW**Trim**

A Power Query step that removes all spaces from the start and end of a text value. 'Trim' does not remove spaces between words — only the extra ones before the first letter and after the last.

WORD TO KNOW**Replace Values**

A Power Query step (right-click a column → Replace Values) that finds every occurrence of a specific value and replaces it with something else. Useful for fixing typos, wrong codes, and inconsistent capitalisation.

To Trim a column: right-click the column header → Transform → Trim. This removes all leading and trailing spaces in that column.

To Replace Values: right-click the column header → Replace Values. Type the value to find (for example 'north') and the replacement (for example 'North'). Click OK.

TRY IT

Right-click the Station Name column header → Transform → Trim. Check that ' Central Square ' has become 'Central Square'.

Right-click the Zone column → Replace Values. Fix 'north' → 'North', then 'EAST' → 'East', then 'west' → 'West'. That is three separate Replace Values steps.

After each Replace Values step, look at the Applied Steps panel. How many steps have been added?

CHECK YOURSELF

Why does Trim not fix the zone capitalisation problem?

Answer: Trim only removes spaces at the start and end of a value. It does not change uppercase or lowercase letters. 'north' with no spaces would not be changed by Trim at all — you need Replace Values for that.

4**Split Column by Delimiter***Separating Period into Quarter and Year***SCENARIO**

Some Period values in the file read "Q1/2024" instead of "Q1 2024".

"I need Quarter and Year as separate columns for the chart," Priya says. "Can you split them?"

Split Column by Delimiter is a Power Query step that divides one column into two (or more) columns based on a character you choose — such as a slash, a comma, or a space. The original column disappears and is replaced by the new columns.

WORD TO KNOW**Delimiter**

A character that separates two pieces of information inside a single field. Common delimiters include: / (slash), , (comma), | (pipe), and a single space. In 'Q1/2024', the slash separates the quarter from the year.

WORD TO KNOW**Split Column**

Home tab (or Transform tab) → Split Column → By Delimiter. Choose the delimiter character, then choose whether to split at every occurrence or just at the first or last one.

After splitting, Power Query creates two columns. Rename them to something meaningful: 'Quarter' (containing Q1, Q2, Q3, Q4) and 'Year' (containing 2024).

TRY IT

Right-click the Period column → Split Column → By Delimiter. Set the delimiter to / (a slash). Choose 'At each occurrence of the delimiter'. Click OK.

Two new columns appear: Period.1 and Period.2. Rename Period.1 to 'Quarter' and Period.2 to 'Year'. Check the data — rows that already had a space ('Q2 2024') will have split differently. How do you fix those?

For the rows that used a space: select the Quarter column, right-click → Replace Values. Replace 'Q2 2024' with 'Q2' (do the same for Q3 2024, Q4 2024). Or: select both columns, add a step to replace the remaining full period strings.

CHECK YOURSELF

After splitting on '/', what will a row that originally read 'Q3 2024' (with a space, not a slash) look like in the new Quarter column?

Answer: It will show the whole value 'Q3 2024' in the Quarter column, with nothing in the Year column — because there was no slash to split on. You need to handle those rows separately with Replace Values.

5**Fill Down***Copying a value through blank cells*

SCENARIO

Three North Bridge records — HC-022, HC-023, and HC-024 — have blank Zone cells. "The Zone should be North for all of them," says Brian. "HC-021 has the right value — can you just copy it down?"

Fill Down copies the value in a cell downwards into every blank cell below it, until it hits another non-blank value. It is used when data was entered only on the first row of a group and the blank rows below it are meant to share the same value.

WORD TO KNOW
Fill Down

A Power Query step that replaces null values in a column with the last non-null value above them. To use it: right-click the column header → Fill → Down.

WORD TO KNOW
null

Power Query's word for a blank or missing value. It is shown as 'null' in the cell, not as an empty space. null values can cause errors in calculations and joins.

To Fill Down: right-click the Zone column header → Fill → Down. Power Query scans from top to bottom. Every null it finds is replaced with the last real value above it.

TRY IT

Scroll to row HC-021 in Power Query. Notice that HC-022, HC-023, and HC-024 show 'null' in the Zone column.

Right-click the Zone column header → Fill → Down. Check rows HC-022, HC-023, and HC-024 again. What value has filled in?

Now check the rows above HC-021. Has any zone value that was already there been affected? Why not?

CHECK YOURSELF

If HC-021 had a null Zone value too, what would Fill Down put into HC-022, HC-023, and HC-024?

Answer: It would continue scanning upward until it found the last non-null Zone value before HC-021. If HC-020 was 'East', then HC-021, HC-022, HC-023, and HC-024 would all get 'East' — which would be wrong. This is why you check the value above before using Fill Down.

6**Remove Duplicates and Filter Out Blank Records***The last two cleaning steps***SCENARIO**

HC-041 is a duplicate of HC-007 — same station, same quarter, slightly different numbers. HC-042 has four blank numeric fields. Including either in the analysis would distort the results. "Get rid of both before you load," Priya says.

Remove Duplicates and Filter Rows are the final clean-up steps. Remove Duplicates keeps the first occurrence of a repeated value and deletes the rest. Filter Rows lets you remove records that match any condition — including records where a specific column is null.

WORD TO KNOW**Remove Duplicates**

A Power Query step that checks for rows with identical values in the selected column(s) and deletes any row that is an exact copy. Only the first occurrence is kept.

WORD TO KNOW**Filter Rows**

The dropdown filter arrow on each column header. Click it to set a condition — for example, 'does not equal null'. Rows that do not match the condition are removed from the query.

To remove duplicates: select the Hire Record ID column, then go to the Home tab → Remove Rows → Remove Duplicates. This keeps one row per ID and removes HC-041.

To filter out HC-042: click the filter arrow on the Total Hires column → Number Filters → does not equal null. This removes any row where Total Hires is blank.

TRY IT

Select the Hire Record ID column. Go to Home → Remove Rows → Remove Duplicates. How many rows remain?

Click the filter arrow on the Total Hires column. Choose Number Filters → is not null (or 'does not equal null'). Check that HC-042 has been removed. How many rows remain now?

Look at the Applied Steps panel. You should now have steps for: Source, Navigation, Changed Type, Renamed Columns, Trimmed Text, Replaced Values (×3), Split Column, Renamed Columns, Fill Down, Removed Duplicates, Filtered Rows. Count them.

CHECK YOURSELF

After removing duplicates and filtering out HC-042, how many records should remain in the dataset?

Answer: 40 records. The file started with 42 records. Removing the duplicate HC-041 brings it to 41. Filtering out HC-042 brings it to 40.

7**Close and Load***Sending the clean data to the spreadsheet***SCENARIO***"The data is clean. Now what?" asks Marcus.**Close & Load sends the clean, transformed version of the data from Power Query into your Excel spreadsheet. The original file is unchanged. If the source data is ever updated, you press Refresh and all your steps run again on the new data.*

Close & Load is the final step. It pushes the transformed data into a new sheet in your workbook. From that sheet, you can build pivot tables, charts, and formulas knowing the data is clean.

WORD TO KNOW**Close & Load**

Home tab → Close & Load. Sends the transformed data from the Power Query Editor into your Excel workbook as a formatted table. The connection to the source data is saved.

WORD TO KNOW**Refresh**

In Excel, right-clicking the loaded table → Refresh (or Data tab → Refresh All) re-runs all your Power Query steps on the latest version of the source data. This is why recording your steps matters.

To close and load: in the Power Query Editor, go to the Home tab → Close & Load. A new sheet will appear in your workbook with a formatted table containing all 40 clean records.

TRY IT

Click Home → Close & Load. A new sheet appears. Count the rows — are there 40 data rows plus one header row?

Click any cell in the loaded table. Go to the Data tab and click Refresh All. Watch the Queries & Connections pane. What happens?

CHECK YOURSELF

What is the benefit of using Close & Load rather than just copying and pasting the cleaned data into a new sheet?

Answer: Close & Load maintains a live connection to the source data and saves all your cleaning steps. If the source data changes, clicking Refresh re-runs every step automatically. A paste is a one-time snapshot — it cannot be refreshed.

Spot the Mistake

Can you see what went wrong?

Marcus cleaned the dataset and loaded it. His Applied Steps panel shows only three steps: Source, Navigation, Changed Type. He is confident the data is ready for analysis.

When Brian builds a summary table by Zone, he finds values called 'north', 'EAST', and 'west' mixed in with 'North', 'East', and 'West'. Two of the Zone values are blank. There are 42 rows in the loaded table, not 40.

WATCH OUT

Marcus changed the data types but skipped Trim, Replace Values, Fill Down, Remove Duplicates, and Filtered Rows. Three steps do not equal a clean dataset. Power Query records only the steps you actually apply.

CHECK YOURSELF

How would you explain to Marcus why having 42 rows instead of 40 is a problem for the analysis?

Answer: HC-041 is a duplicate — it counts Victoria Park Q3 2024 twice, inflating hires and revenue for that station. HC-042 has blank numeric fields — it would cause errors in averages and totals. Any calculation including those rows would give wrong answers.

Mini-Mission

Two minutes — prove it

Load Pack3_GW_data.xlsx into Power Query. Without referring back to the chunks, apply all the cleaning steps you can remember. Your target: a loaded table with exactly 40 rows, no spaces in Station Name, consistent Zone capitalisation, no blank zones, Quarter and Year as separate columns, no duplicates, no blank records.

When you are done, count the Applied Steps. You should have at least ten.

You Can Now Answer

These are the kinds of questions that come up in Scenario Demonstration 2

You should now be able to answer all of the following:

- ☐ What is Power Query and why do we use it instead of editing data directly in Excel?
- ☐ What does the Applied Steps panel show, and why does it matter?
- ☐ How do you remove leading and trailing spaces from a text column?
- ☐ How do you fix inconsistent capitalisation across a whole column?
- ☐ How do you split a column that contains two pieces of information separated by a slash?

- ☐ What is Fill Down and when do you use it?
- ☐ How do you remove duplicate records in Power Query?
- ☐ What does Close & Load do, and what is the benefit of using Refresh?

If You Got Stuck

No problem — here is what to practise next

Power Query feels strange the first time. Every data analyst spent a while just clicking around and undoing steps. The point of recording steps is that mistakes are undoable — just delete the step from the Applied Steps panel and try again.

— Priya, GreenWheels Data Lead

If Chunk 4 (Split by Delimiter) felt tricky: open a fresh copy of the file, load into Power Query, and practice the split three times until the steps feel automatic. Then try it on the Period column with a comma instead of a slash — see what changes.

If Chunk 5 (Fill Down) felt uncertain: try deliberately leaving a value at the top of a column blank before you fill down, and observe what happens. Understanding what Fill Down does when the first value is blank is important.

Ready for more? Pack 4 builds on this clean data — adding calculated columns, performance bands, and group summaries using more advanced Power Query features.